

Part 1(a): Default vs. ideal hyperparameters

d8, compute-matched at 614M tokens. Default run = 2.927 (seed noise: sd 0.003 across seeds 42/1/2). 'Ideal' = best value found in the one-at-a-time sweep, with later follow-ups where they refine it.

Hyperparameter	Default	Ideal (best found)	Loss at ideal	Gain vs default	Confidence	Notes
Learning rate	3e-3	3e-3 (keep)	2.927	0	High (clear U-shape)	Flat from 1.5e-3 to 6e-3 (± 0.02); too low hurts more than too high. With 4% warmup the best LR rises to 6e-3 (1(b)).
Batch size	64	32	2.919 (3-seed mean 2.920)	-0.006	High: better at all 3 seeds	Smaller is better down to 32 (more steps for the same tokens); 16 is no better; ≥ 128 hurts sharply at LR 3e-3.
Warmup (% of steps)	1%	$\geq 4\%$ (best tested: 16%)	4%: 2.924 · 16%: 2.917	-0.003 to -0.010	Medium: monotone trend, single seeds	No optimum up to 16%; beyond ~1% it's a flat, safe plateau. Matters far more at high LR.
Weight decay	0.1	0.2	2.919 (3-seed mean 2.919)	-0.008	High: better at all 3 seeds	Shallow optimum: 0.4 $\rightarrow$ 2.922, 0.05 $\rightarrow$ 2.937. Acts via weight-norm control, not regularization (single epoch).

Recommended d8 recipe

Recipe	Changes from default	Result	Status
Default	—	2.927 (seeds: 2.924-2.930)	baseline
1(a) winners, one at a time	bs 32 <i>or</i> wd 0.2 <i>or</i> warmup 4-16%	2.917-2.924 each	each measured alone
Tuned recipe (tested jointly)	$\beta 2$ 0.99, warmup 4%, wd 0.2	2.915 (d8, 2(b))	measured together, one seed
High-LR recipe (tested jointly)	LR 6e-3, warmup 4%	2.916 (1(b))	measured together, one seed
Suggested next test	bs 32, wd 0.2, warmup 4%, $\beta 2$ 0.99 (LR 3e-3)	not yet run	gains may not add up; run 2-3 seeds

How to read this. The default is already close to optimal: every single-hyperparameter improvement is 0.003-0.010, about 1-3 seed standard deviations, and the best jointly-tested recipe gains 0.012. Learning rate and batch size matter most for *avoiding* bad settings (spreads of 0.08-0.09 across the sweep), while warmup and weight decay give small, safe gains. The two replicated wins (bs 32, wd 0.2) are the ones to trust; warmup's gain is a consistent trend but each point is a single seed.